

- Write a Python function that accepts a string and counts the number of upper and lowercase letters. *Sample String*: 'The quick Brown Fox'
Expected Output: No. of uppercase characters : 3
No. of Lower-case Characters: 12
- Write a Python function that takes a list and returns a new list with distinct elements from the first list. *Sample List* : [1,2,3,3,3,3,4,5]
Unique List : [1, 2, 3, 4, 5]
- Write a Python function that checks whether a passed string is a palindrome or not.
- Write a Python program to square the elements of a list using the map() function.
- Write a Python program to add two given lists and find the difference between them. Use the map() function.
input:[6, 5, 3, 9][0, 1, 7, 7]Result:[(6, 6), (6, 4), (10, -4), (16, 2)]
- Write a Python program to create a lambda function that adds 15 to a given number passed in as an argument, also create a lambda function that multiplies argument x with argument y and prints the result.

Sample Output:

25

48

- Write a Python program to sort a list of tuples using Lambda.

Original list of tuples:

[('English', 88), ('Science', 90), ('Maths', 97), ('Social sciences', 82)]

Sorting the List of Tuples:

[('Social sciences', 82), ('English', 88), ('Science', 90), ('Maths', 97)]

- Write a Python program to remove None values from a given list using the lambda function. Original list:

[12, 0, None, 23, None, -55, 234, 89, None, 0, 6, -12]

Remove None value from the said list:

[12, 0, 23, -55, 234, 89, 0, 6, -12]

- Write a Python function to check whether a number is "Perfect" or not. In number theory, a perfect number is a positive integer that is equal to the sum of its proper positive divisors, that is, the sum of its positive divisors excluding the number itself (also known as its aliquot sum). Equivalently, a perfect number is a number that is half the sum of all of its positive divisors (including itself)
- Write a Python program to listify the list of given strings individually using Python map. Original list: ['Red', 'Blue', 'Black', 'White', 'Pink'] After listify the list of strings are: [['R', 'e', 'd'], ['B', 'l', 'u', 'e'], ['B', 'l', 'a', 'c', 'k'], ['W', 'h', 'i', 't', 'e'], ['P', 'i', 'n', 'k']]
- Write a Python program to create a function that takes one argument, and that argument will be multiplied with an unknown given number. Sample Output:

Double the number of 15 = 30

Triple the number of 15 = 45

Quadruple the number of 15 = 60

Quintuple the number 15 = 75

- Write a Python program that creates a list of names and uses the filter function to extract names that start with a vowel (A, E, I, O, U).
- Write a Python program that creates a list of tuples, each containing a city name and its population. Use the filter function to extract cities with a population over 2 million.

Regular Expressions:

- Write a Python program to find all three, four, and five-character words in a string.
- Write a Python script to transform a camelCase identifier into snake_case and then print the result.
- Write a Python program to find URLs in a string.
- Write a Python program to remove everything except alphanumeric characters from a string.
- Write a Python program to split a string with multiple delimiters.
- Write a Python program to replace whitespaces with an underscore and vice versa.
- Write a Python program that matches a string that has an *a* followed by zero or more *b*'s.
- Write a Python program that matches a string that has an *a* followed by zero or one *'b'*.
- Write a Python program that matches a string with an *a* followed by two to three *'b'*.
- Write a Python program that matches a string with an *a* ' followed by anything ending in *'b'*.
- Write a Python program that matches a word containing *'z'*, not the start or end of the word.

- Write a Python program that starts each string with a specific number.
- Write a Python program to get the year, month, and date from a URL.
- Write a Python program to separate and print the numbers and their position in a given string.
- Write a Python program to replace with given number of occurrences of space, comma, or dot with a colon.